"Physics as Information Processing" ~ Ander Aguirre ~ Discussion 1

Source: YouTube **Playlist**: Physics as Information Processing ~ Chris Fields ~ 2023 **Duration**: 1:06:28 | **Uploaded**: Unknown **Transcript method**: auto_caption

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hello and welcome everyone it's June 3rd 2023 we're in the first discussion section for physics as information processing course with Chris fields so welcome to all who are here on and off camera really exciting to see so many people joining for this participatory discussion section so in this first interval we're going to go to anyone who raises their hand and it would be awesome to hear from everyone who'd like to share what excites them about what they heard in the first lecture how has their background and journey led to them joining what are they hoping to learn and do in this course so first hand we'll get to go maybe Andre you can actually start first just speaking as a normal participant and then meanwhile anyone is welcome to raise their hands that sounds good so I I am a mathematician by training um but I've always been interested in the more physics side of things and in particular you know I followed Chris's work I'll work pretty closely since probably uh and I don't know I'm just very interested in the whole ontology of quantum measurement what it means um I feel like you know very often when it's taught in physics classes it's hand waved away and even though I myself uh subscribe to the philosophy of you know shut up and compute sometimes you know not everything you need to do a little bit of philosophy maybe and this may be one of the things where you need to do it I usually to address it from uh first principles right and I think Chris is doing that so I'm just here to learn more and yeah that's basically it awesome I'll just go from the top left because I'm not necessarily watching like what order but Ross first hi um I'm interested in the kind of overlap between cognitive science and physics and seeing if uh there are any pan psychist implications of the observation that uh physical processes seem to follow the same kind of principle that we see apply to uh brains and cognitive systems awesome thank you Avo hi I am uh I'm uh mixed backgrounds uh mostly physics and cognitive science right now I could be qualified as a philosopher of physics and cognitive science and I'm interested in uh basically formalizing participatorialism the idea that Observer participate in creating reality I did some work in formalizing that in the case of social norms and things like language and I met a block based on my lack of knowledge of quantum physics and the mathematical tools that I used for that so I'm here in a course to get those formal skills awesome thank you evil blue I think Alex was first okay Alexi go for it oh sorry hi I'm Alexia I'm a clinical psychologist in Maryland in the states I uh work with patients do Psychotherapy mostly my interest came through uh Mark zolmes uh who is uh you know the Creator and leader of neuroscycle analysis and he has been collaborating with friston uh on uh heart problem of Consciousness and he's not working on AI with emotions and so uh he you know I blame him I say it with respect and love that for uh introducing mathematics into our field which was mostly devoid of mathematics and uh people started looking at these partial differential equations and Mark with blankets uh so I wrote a paper about a possible extension of their model uh to include chaos theory and I just want to understand deeper uh Concepts involved I have some understanding of them but perhaps not enough and I'm enjoying this course a great deal I mean I think Chris among other things is a fantastic teacher and he is able to express things very clearly and it just I have oh I have Goosebumps so I'm very grateful to everybody and thank you so much cool blue then Francesco hi everyone I'm blue Knight I'm with the active inference Institute so I've been tracking this work of Chris's um you know connecting Quantum information and active inference for I don't know a couple couple years now through through that thread um I'm super interested actually in um the aspect of space-time arising from communication I think that that's like fascinating to dive into and how it connects to the like Buddhist and maybe more Eastern religion like concept of dependent arising like how like the interdependence of all things leads to their existence um and and so from from that aspect I think that that's like the what I'm most interested in exploring my background is um Neuroscience but I've also been studying practicing Buddhism for like 25 years so um yeah that's it thanks awesome Francesco then Corby thanks Daniel ciao everyone super happy to be here I also have a mixed background because I studied cognitive anthropology and philosophy of science and currently a few months ago I started a PhD in artificial intelligence in an es application in the educational field and yes I'm actually super excited about the all the science implications that Chris Fields mentioned in the first lecture in connection I have no background in physics but the connections between the first principles of physics and free energy principle and scientific condition in general is what is really getting me excited in these days thank you cool Corby then anyone else who raises their hint uh yeah great to be here um my background is in Quantum engineering uh I've always been motivated by figuring out what the next big thing in Computing was in high school Morris law was sort of coming to an end and it seemed like a really big opportunity to figure out what the next big Computing architecture might be so I did academic research and no more for computing Quantum Computing and Quantum sensors and I felt like I was a little bit too early for all of those things so I went into uh cloud and cloud gaming and streaming AI services but I'm still extremely motivated to figure out what the next big thing is in the holographic principle seems like a good foundation to explore that awesome I'll raise my own hand than anyone else is uh welcome to go so I'm Daniel I'm a researcher at The Institute and have known under for some years and learns about all kinds of stochastic matrices and how the math and the quantum were linked and we had talked and joked about learning from Chris fields for a long time so when the opportunity arose to enact it and extend it and open it we were very excited and also don't have much of a formal background in physics outside of working with Jacob and Ali on Livestream 49. so it's just going to be a great time to learn and see where it all goes anyone else want to say uh hello in this kind of opening section okay well continuing forward of course please just feel free to raise your hand anytime you'd like to jump into the stack under for for several minutes or however you see could you reflect on lecture one where did Chris enter how did he proceed and where did the lecture take us thank you yeah so can you zoom in a little bit more on the PDF can you hear me thank you good go for it let's uh take a look at the slides from last time so I think if if we were to summarize this um whole lecture basically what Chris was trying to say is that there are many hints coming from different directions pointing at the same thing right pointing at the fact that you know information no will have a central aspect uh in physics in the future but also possibly biology right so the earliest games came from statistical mechanics if you wish but it also uh quantum mechanics right again we need to make a like an important but subtle distinction between what it is the uncertainty principle right because you know we think of that as as the informational part right like a limit on the resolution on information uh and what is the observer effect just to make that distinction clear I think Chris is talking about the observer effect the uncertainty principle is a an artifact of the wave nature of you know quantum mechanics right you can you can get it from properties of the Fourier transform with cautious Schwartz inequality but at the end of the day it's what I was saying earlier right what is the basic picture of measurement right and that's sort of increase with glazebrook and Marciano is trying to address and the first two hints came from from quantum mechanics around 1900 and statistical mechanics a few decades earlier then there's you know more stuff right like uh obviously businesses uh resisted to inertia of hundreds of years of a totally objective you know Observer independent universe but the hints continued to pile on some stuff coming from math but perhaps the most interesting is is again you know how all those paradoxes from statistical mechanics were were made sense of with Land hours proposal and then as physics got more fancy past you know qft and stuff you know people started thinking about Quantum information more seriously so we had Dell serum and all these proportions by Deutsch and Wheeler and so on and I and I should say that um this is not a fringe idea in physics uh the Paradigm of it from qubit is very much mainstream I mean Google it a lot of physicists think about it you know I don't know how many of them you know talk about it openly but you know it's it's been a widespread folk intuition so to speak that you know information will have this sort of central role yeah and the last sort of hints pointing in that direction of information having a control what came on the one side from high energy stuff and that's the holographic principle is it originally arose in the context of black hole for Dynamics but then you know it was extended to you to you know what's called the abs-cft correspondence and the very most recent stuff was you know it's Basics on the one side from high energy staff sorry I can't speak a little more loudly it's all good continue honor yeah I thought there was a question uh someone was just unmuted it's all good okay uh yeah so so that's basically it right so let's there's hints from many directions and most recently uh you know the work of Tristan and Levin coming from biology all pointing in the same direction right that there we might have to make sense of some sort of potentially scale free information processing uh Peridot or for what it means to be an observer you know sort of all of these things are pointing in that direction and I think Chris's work is is addressing that head-on right as I said earlier you know sometimes you know it's it's good to do computations but sometimes you need to take a step back and think more philosophically about what you're doing and what the bigger picture is and then maybe after people talk a little bit I can you know we can warm up to the second lecture we can the other windows that I have open here um some of them are the papers that are to come yeah anyone please feel free to um raise your hand on this Alexia than anyone else well I had one uh thought about this thing you know when you mentioned uh that there's various perspectives from the same concept and I'm trying to be cautious on the one hand I think again this amazing Clarity and precision with which Chris expressed um uh thoughts is is fantastic and enjoyable on the other hand it requires simplification and generalization and what I observe in my field is take people take a term like entropy and run with it and then they get very quickly from thermodynamic entropy to Shannon to you know whatever entropy and they talk about sort of um and that's not the case I mean if we go back to class use and all the others there was a system of postulates and axioms that must be maintained and if you step outside of that so like us organisms we're open systems we're not closed systems so we should not apply directly second world thermodynamics and you know I I deal a little more with kolmogorov's and I entropy and which is not the same thing and I guess when we very quickly we move from sort of thermodynamics to you know uh to statistical mechanics with boltzmann to you know all the way around and we assume it is the same thing I think we may need to be careful because we are just ignoring the differences in assumptions you know Shannon talked about Telegraph he talked about a receiver getting a message from a sender right and clausius talked about you know uh gas so um but but I do think it's a balancing act and there's no way to not have simplification and generalization I just wanted to say somewhere somehow we need to kind of you know go back there and just say you know uh entropy is not entropy is not entropy and all these things are kind of you know long papers right yes Jakob then Ali than anyone else I um I just wanted to uh ask for I guess perhaps uh slightly longer comment from Alexian that what you see as the fundamental differences in different descriptions of entropy and whether there is some I guess General concept that could be extracted from all of them such as level of information or uncertainty about a particular system even though these specific uh formulations of entropy they talk about they were derived from different specific systems and they have different equations describing them and they mean slightly different things do um do you think it would be wrong to say that there is some commonality between them did I answer Daniel or yeah I don't know I just think we need to at some point formalize it and um and be careful with these generalizations for example uh when we talk about active inference and Tristan right you know when we say living organisms must minimize I'm quoting first and by the way these are not my thoughts the entropy of sensory States right and then people take that idea sometimes and generalize and they say that that's what life is minimization of entropy right but if you actually look into a human brain and measure entropy then it goes all over the place up down and in between uh and so I I don't think that's an accurate description and the reason for that is we're we shifted away from the uh assumptions and uh again so comma version I entropy talks about the phase space and the Divergence of trajectories in face space and sort of the volume in face space while when you talk about this for instance uh entropy of sensory States these are probability distributions and Shannon talked about the uncertainty of the receiver about the message sent by the sender so if you want to extract the same thing I suppose you can but but it's important not to lose at least the difference between what kind of object are we applying to are we looking at probability distributions are we looking at Phase space are we looking at ideal gas where are we at and we're just not to say entropies entropies entropy that's not the case I mean it's just a formula but depending on the context assumptions and and things it can be very different thanks Ollie and other uh yes in regard to Alex's comments I just wanted to mention the research of people like Rudolph panel and Stefan Cerner because uh for many years they've been exploring this idea of different conceptions of entropy and they've come up with at least a dozen different meanings for entropy and they've formulated it in a kind of generalized framework but at the same time they've also devised uh a way to somehow classify them into equivalence classes and to see whether one or more of those conceptions of entropy can be theoretically derived from uh from some others or they're totally distinct and different um conceptions of entropy so yes I agree that in some situations we need to be more specific about what exactly we mean by entropy and information uh I mean based on the question or the situation of Interest thanks aval ating the entrepreneur discussion I want to emphasize that entrepates plnp so it's a function for probability distribution which is uh I think I'm not sure what the term is English but it's a measure of how big something is in a space and to get that you have to define a space and in many cases it's preservatory to decide what the space you're looking at is for example you talk I don't know if it was like say it was like say someone talked about uh entropy of fmri well those are aggregate measurements of General activity this is not the space of all possible configuration and even if I add something like the space of all and to say functional configuration the state in your own that reflect activity this would be another thing than the space of I don't know Atomic configuration where the specific atom are so yeah you don't have a warranty that's when you define entropy you define it on a main full space that stopped given so let's I kind of think there is one entropy and that's it and need some info constructs but also we have to if we agree to that then we can tell that we have to find it it's not just given out there in the data well certainly being precise in the formalism and in natural language for what we mean by information and entropy and all these terms is pretty important I mean the course we're in is physics as information processing but without an understanding of how information is being meant here or what measurements and Transformations were were performing what informational operations then it it is it a non-sequitur so how do we how do we go about um clarifying or understanding what kinds of theoretical results in a given domain or like what is shown on the timeline here might apply to some other area Just one thought anyone else can raise their hands with uh a thought or a question otherwise Andre you can um maybe flip over to to select a few of the submitted questions cool could you zoom in um with control plus a fair amount thanks so everyone um we really appreciate the submitted questions we've been able to get 13 answered by Chris so far and um as with the transcripts of the discussions and lectures the questions and answers are going to be part of what is published by the octave Journal so we really encourage basic questions Advanced questions just there's no question that doesn't make sense to submit so under maybe just pick one question that you think's interesting to begin with and summarize uh Chris's response and then everyone else is welcome to kind of add some other thoughts on this no so I'm going to scroll up and down just to sort of sample uh let me start with the first two can you hear me well yep uh since perhaps I think you submitted this as a trial right is that is that right yeah so what is information about this time here's Chris's response I guess everyone has a chance to look at it now the one if I am to summarize this the one thing I've heard that you know it's personally the one I like the most um the answer at least as an adage to summarize it is that information um are differences that make a difference this I might have heard from Chris I might have heard from someone else I don't remember but that's one thing now on to the question of what is time and I guess this also addresses uh Alex's question of what is entropy I so I'll make a bigger point now uh what is time this is partially addressed and I guess the main thing I'll be doing here is rather than look back look forward into what's to come in the course in particular I'm thinking about you know this these two papers uh and in particular [Music] particular and I finally chat with with Chris about this uh this picture over here okay more yeah I'll get back to you in a bit but basically this this also sort of like kills two birds with a stone right this picture and especially the text around it and the title of this paper is the physical meaning of the holographic principle okay you guys can see here archive 2020 2210 blah blah blah anyways this addresses the notion of you know the time on entropy because the the point that's the claim that's made in this paper is that these are local Observer relative definitions and we can look into more detail around it but basically the coordinate ta time is an entropic measure so it is non-decreasing and it goes up with The Observer measuring uh entropy in the system because of the symmetry of the definition the second law applies to both so both of server and environment C entropy increasing in their environment and vice versa anyways from from this picture one gets that you have this internal Circle time qrf so this just is is a necessary artifact of recording information into a memory that's that's the claim made in the paper and that essentially induces the passenger of time at least from the perspective of the Observer um it also I think Alexi was talking about which I totally uh understand and agree with different Notions of of entropy now I suppose as the the you know I I would be the one to to like definitions and so on um but I but

I do think it's it's it's good to to have a certain amount of flexibility um so I see the same as energy right like there are different kinds of energy you know there's energy mass equivalences you know if you push it into Super symmetry you can think of it as equivalent as Force also but I I do think there is some value in being sloppy or intuitive about about the notion so so the fact that you know entropy yes there are different definitions I I don't I don't see it quite as a problem you know one can explore here around this this page there is a definition of you know local entropic time and yeah one can think of it as you know face Pace or what can think of it as you know coding Theory but it is at the end of the day I do think it is true that both of them are hinting at the same direction just like you know you may have different kinds of energy but um you know they're not unrelated Concepts and and yeah I hope one day we get very precise definitions about everything but but for now I think it's good to speculate okay there's a question yeah Corby and then anyone else on this yeah quick question so I haven't read that specific paper but is there any work on the frequency or the rate at which one could read and write information on the boundary yes well I I don't I don't I can't think of the the paper exactly but this is lower bounded I think Chris talked about it if not in the last lecture I I think I've seen a talk of him on YouTube it's lower bounded by the uncertainty principle right um so so for instance uh plants Planck's constant is in units of action which is energy times time right so Leonard's principle says you need a certain amount of energy you know to raise information you have time right and then you have an energy time uncertainty principle putting those things together right you have any minimum amount of action which is you know literally the minimum uh the action is measuring what I said earlier right like differences that make a difference the minimum difference that you can make so the minimum difference to you know you raise a bit or rewrite it on the boundary so to speak right uh so yeah that would be given by the uncertain relation energy times time it's I don't know exactly the figure is but it's obviously very small compared to Everyday Life got it and with is there work on the upper bound like for example would it be the number of degrees of freedom on Boundary no I don't think it's bonded from above but but obviously you know then we do want to buy it from above for technological reasons if you wish but I I think that is a an engineering problem not not one of you know physics um under a few few questions on this figure what are y and e B the blue oval as Chris has pointed out is going to consistently be the screen the blanket and so we often associate that with uh at least some blankets include these information gathering utilizing systems so what is happening internal to the agent here what kind of computation is happening and is this something that is actually proposed to be occurring or is this a schematic or a map of how we could think about something occurring I think this question is going to be better addressed by Chris himself probably in the next lecture um for now what I can say is I can reiterate what I was trying to say earlier which is that with these addresses and you know these basic measurement pictures so to speak is the observer effect right like this notion that you cannot observe something we thought without disturbing the environment so if you then try to make sense of this by constraining you know from first principles Land hours principle and so on uh basic requirements on information processing from the point of view of energy then you get that this boundary is constrained in some ways right this Y is What's called the memory sector meaning that you don't just act on on the immediate information that you perceive you may also act on you know start that story that you have perceived before and then sure e the red part is what you're perceiving right now the environment uh but actually I don't know if I have it in here uh but this is further broken up into other parts so not schematically here let me try to find it over here no I don't know actually bear with me for a second I think you might be here if that's the same picture anyways I I'm not I don't want to waste time by trying to find it but back to the original picture yeah sure you use what you're measuring and again Chris will talk about this in more detail yeah but from first principle assumptions they deduce that you're gonna have to you know deploy your finite resources for gathering information in clever ways on the boundary right so you're gonna have to deploy some resources for memory some resources uh as a free energy uh as a heat sink right so you're going to be processing information so you're going to be dumping you know so-called low quality stuff into the environment you're going to be dumping Heat and then other than the memory you're going to be focusing on what's going on right now in this environment and what I wanted to say is that it can be further broken up into what's called pointer and reference sectors so you can imagine this e-sector here has to be broken into further sectors because then there's the other first principle notion okay so what does it mean to observe some something the differences that make a difference right so you can only there's only information with context right there is no information per se it's all semantics right so you've only observe things in context so some things are going to have to stay the same namely you know reference States here and some things are going to Bear right so you can think of this as the gauge in your car right obviously the circle stays the same right and that's your context so to speak that's the with the you know numbers in it they never go away they stay the same but as you speed up around the gauge is going to change and that's how you know only in that context you you can tell you're on speed when you're driving a car so anyways this is just a teacher for things to come you can go back to more questions then what do you think I'll pick maybe two two or three more over here yeah pick another question or anyone raising their hand or if anyone in the live chat wants to write something I'll read it how about the wick rotation could you summarize what the wick rotation is and then State this question about it so yeah that's a great question I knew the answer but I wish I knew the answer was better but unfortunately speaking um the weak rotation is just so is Chris said you know you take a quantity you multiply by I square root of negative one that amounts to a 90 degree rotation in a complex plane fine now that appears to be a trick and what it does effectively is take you from quantum mechanics to statistical mechanics roughly speaking now why it works that's probably I mean I have no clue I don't know the answer myself but now when things appear to be a trick and they get you something maybe there's more to it than than what's you know Apparent at first uh but I myself don't know to know the answer it's a very good question uh but I I do think it may have to do with the fact that on the one hand Quantum and statistical mechanics both are talking about constraints on information processing maybe from different perspectives but again that's just the speculation on my I I don't know so it's maybe a way of relating the two did we just review just read Chris's full answer to this read the question and then Chris's full answer foreign space and then I'll I'll just read Chris's answer then anyone please raise your hand if you want to like add something more on this he wrote in a sense yes time is perpendicular to our three dimensions of space matter and energy we ourselves exist in time Wick is pointing to the intimate relationship between how we measure time and how we determine or judge that something maintains its identity Through Time IE stays the same thing through a time to tell time by a clock I have to be sure I'm looking at the same clock set the same way Etc we'll get to this in July and August could you connect this under to some of the um semantic information flows discussed in live stream 17 or any of the other uh cone cocone diagrams from the papers that you showed earlier uh sure again I have to preface it by saying that I am working on understanding it myself um just from you know small Email exchange that I had with Chris and website in this paper one can sort of see that it's related to the picture I was showing earlier let me this one right but let's scroll down a little bit to page to page 23 24 here they took out the rotation again I think it has it has to do with uh at least from what I heard from Chris gay email has to this notion of local local time the definition of local time in this observation picture and you know the fact that you know rotate 90 degrees twice you get the opposite sign basically right so uh the local time effectively that's going to ensure that that the picture is symmetric and that the flow is going to be opposite in in the direction uh towards the The Observer and towards the environment okay Alexi then anyone else raising their hands I I am struggling to understand and part of my difficulty bear with me ever being so uh kind of but I remember Sean Carroll's uh lecture on time in different versions of physics and it took a whole hour so are we talking Newtonian time which is universally the same I would talk in relative time uh you know philosophically are we talking about people who believe that only present here and now is real while future and past do not exist you know so when we say time is perpendicular too my my sense is that just like Chris mentioned and what you were Reinventing the time we're creating a new definition by calling it the same word we're all familiar with and that creates some dissonance where I I I just think that maybe we need to kind of say that this is something new and if it's not something new then which version of the time are we talking about Einstein Newtonian or whatever you know it's a simple question if we before we say it's perpendicular do people agree that future is uh real or not you know again speculation um neither here nor then if we take Chris seriously that communication is an ontological primary and that rather than putting Communications protocols in some sort of time Newtonian time Einstein time these are these different time Concepts these different scaffolds that then communication gets inscribed within or stretched within but if we start with a topological understanding of communication rather than a geometric understanding of space-time so the topology of the blanket and the observer in their sensorium then different kinds of cognitive agents ranging from what we would call non-living to what we could call living they depending on their cognitive sophistication are able to create or project different time Concepts time Concepts that are not useful will lead to their own material dissolution the failure to maintain that kind of cognitive sophistication which requires um not just information gathering utilization but also energy Gathering and utilization and so when you have multiple agents who are um whether you conceptualize them as within a message passing relationship and having uh no surprise on the clock on the wall then they're like in a local time synchrony but the primary case is one entity engaging with its sensorium and its self-modeling time in terms of progression and so these previous time Concepts and these different attempts to kind of survey the structure of space-time reflect different cognitive models that cognitive agents have proposed for time and that communication actually like takes ontological Primacy over all of these ramified um potentially overlapping potentially non-overlapping time Concepts but their Downstream of Agents cognitive models not vice versa I'm not sure it's how I interpreted what Chris said about communication being an ontological primary though so if we take one instance of a covid-19 virus a single kind of instance right it's got no memory and no ability to plan is there a local time for that virus versus you know Humanity observing how the virus spreads across the globe and you know we have agents with memory and planning you know so is there a time relative to a single Photon you know I I just I mean it's it's my hunch is that there's a possibility that we're simply inventing something brand new and defining it brand new and I just they draw in parallel back to all these other times you know I understand the effort but it's it's just is it is that a brand new time is it some other kind of time I don't know so I I understand Alex's question uh and I think it's sort of related to at least morally speaking to your concern about different kinds of entropy right and what I can say is that also we have Gladys Brook uh Marciano and so on they have this um recent paper on Quantum error correction uh so uh I have yet to read it and take a closer look but here they talk about space time right and my punch to address Alex's question is that if you could sort of recover the more familiar Notions of time in a correspondence principle sort of way you know by taking some you know Newtonian limit or H bar goes to zero or speed of light was infinity and so on uh that would be very satisfactory sure they may be doing a new thing but you know recording old stuff and then you think it's a good sanity check and so long as we can see that being worked out I think I'd be very happy and just to the virus point I would argue that viruses through their physical engagement they do have memory but they don't necessarily have the awareness or a metacognition on that memory except as granted by intergenerational evolutionary type pressure but they do have past events modifying them within some set of possibilities they're just being them and their internal time is is on there isn't a cognitive Observer necessarily in terms of like a narrative I I am watching the clock but they do change but it's certainly very provocative if this is not or we could also look to Kronos and Kairos and about the the relationship of time and timeliness and agentic time so I think there's and then at this point are we stretching just this one four-letter word too far just like we brought up with information entropy I mean can we think about these as kind of like continents and then just if someone said I visited this continent you know that you need to specify a lot more or what granularity are these topics Corby and then anyone else raising their hands yeah on this topic before a Markov blanket you have a transition Matrix my question is how do you how are you allowed to increment that transition Matrix to calculate a future State like what does that mean in terms of time uh are you only allowed to increment it by like the lower bound time which is H over oh and KBT does that question make sense I'll give a first pass on this we have discrete time and continuous time models in active inference for a discrete time model where you have the B transition Matrix then the choice of delta T is a modeler's decision and we talked about this a lot in live stream 52 on accelerated optimization there's some delta T that's like going to get you the most accelerated optimization if you have too small of delta T then you're either oversampling or you're maybe even physically unrealistic like you pointed to or if you have some long delta T you're also not making an informative simulation in the discrete time case where you're explicitly making time steps contrast with the continuous time generative model where we're doing more of a Taylor series expansion on the generalized coordinates at which point delta T is not required you're merely just taking an extended approximation from a given snapshot and so delta T doesn't even come into play got it okay that makes sense I don't know how it ties into time but it helps clarify a little bit okay anyone else can raise their hand or um Andre kind of maybe one more question or one more topic from a paper just what's one more puzzle piece that you think gives us a first coat of paints from questions submitted or from one of these papers that you have up what's something that that helps us connect the dots a little bit uh so I don't I don't think I have a good answer let me look at the questions again but regarding time I'll just reiterate uh what I said I say record these questions which is that I believe Chris will address it in in this bit I I don't know if you can see what I'm sharing on the screen but the paper will start all these communication protocols blah blah blah from March of this year uh does talk about space-time it's a Quantum uh error correction code and to me personally it'd be nice to see if that can be if more familiar Notions of time such as special relativity can be recovered past limits in correspondence principles yeah do you want me to say anything else if anyone could even just summarize like what what would these what are these codes referring to or how are are these codes one and the same as communication protocols or are they related to the blanket and the the heat that was being output as waste What will what will these codes allow us to do better if we understood them more I don't know that I can answer that right now I don't I don't think they understand the question um yeah I I just don't know sorry Terry and then anyone else raising their hands um I I'm not a physicist I don't understand maths at all M but what attracted me to this um idea of physics is information processing is that it feels like at a base level the whatever the X on normal external status it could be perceived as made up of information you know we we kind of think of ourselves living in a particular universe but you know at a base level there are these and stop me if I'm just talking 14 year old nonsense here but you know that there's there's no Quantum Fields they feel like informational fields the the the concept of of physics is purely information that we then interpret as a physical Universe which we inhabit because we can't see beyond our Markov blanket I find this a really exciting idea that as a that whenever you start to think of the um surprisal as the um variational free energy between you know sort of your your prior and your posterior if you start to think of that variational free energy not as energy but as an information Gap so that if you see the energy as actual you know because energy when you think of energy in physics terms it's kind of this magical thing that does that allows work to happen or you know but but what is it you know we've got a a term for it and we've got we've got we've got a you know we there's a jewels you know or whatever you know but but what does it mean it has this magical property but if you think of it as an information um then it starts to make sense so variational free energy starts to be seen as a difference in information between what you uh your your prior and your exterior so whenever you start to think about time time is this conceptual metric that we use almost because we have memory and um I I wonder if we are getting uh trapped in this semantics and all I have is English because I don't understand the maths really but um we we you know you guys come in here and you do understand the maths and then you convert it into English which I think I understand and then I can think of it like a kid in a sweaty shop or a physics class at the 14 years of age I don't really know what I'm saying I don't know what I'm saying but at a fundamental level everything that we're talking about is in for I I I can understand this as information and I'm excited to see how things develop what are your thoughts on this craziness so many uh great pieces in there physics as information processing what is it that actually allows the steam engine to run and the ball to move that kind of physics and then as the timeline laid out also occurring during a period of immense technology and and apparatus development and and digitization all of this and unconventional Computing as well it's almost like this uh materialist dualism that instead of with with mind and body that there was some kind of dualism between the types of forces and energies and ways of thinking about what made the steam engine work and then what makes a computer work and so you'd understand how the computer could fall off a cliff but then you wouldn't necessarily be able to use that exact kind of mechanics and physics and Dynamics to study like how the computer operated but you could say it weighed 11 pounds and so I think the the provocation and then which is as you hinted towards um first off would be awesome to have a natural language math and graphical synthesis the triple play but also then to use a common approach to understanding this and what wasn't working or is not proposed to work is trying to fit into that non-cognitive universe but rather if we take the cognitive Primacy of communication then all of these forces as inferred by the sensorium which is all we're ever going to have totally the most valid first principles constraint to make maybe then we'll be able to have a lot more synthesis about even how steam engines work when we understand that word Bound in that inference by our sensorium and all that comes along with that in the variational free energy bounding surprise on our generative model um Dean um I'm gonna I'm gonna come at this from the position of a 14 year old lens because I don't think there's anything unique about that um I think on most basic level what I took away from Chris's lecture was that there are that there's information processing of at least two types uh one is figuring it out as in figure five or any of these other figures that are up on the screen right now or anything that stabilizes the world around us I think examples have come up in this conversation already of apology or geometry there's a second type of information processing which is outing the figure type of information which is searching and I think Lexi used the expression which time are we talking about so these two types of processes are quite different if you do a comparative analysis of the two you have information practicing and you have to have both of the processes to really get a sense conceptually of what it is the communication is really talking about if you're our information practicing your way finding until you collapse to something we might describe in a verb tense as wave found it's in the quantizing of that invariance state that vast ocean if you want to think of it metaphorically oh to all all Horizons on the manifold into a variant or a nuanced or a discretized or an orientation type of State where all of a sudden things like currents and temperature ships and even think things that we've applied labels to like entropy and a transition Matrix suddenly appear out of the seam and nothingness or the invariance so that's not the math but that's kind of stepping back and saying what is the math trying to say I think it boils down to the comparative analysis and then I think I'm hoping that in the coming lectures Chris starts opening that up and doing kind of the back and forth between the two types of of figuring it out and Outing the figures because I think when both are are part of the conversation that's when we start to get a sense of what's really going on in this sort of closing sessions especially if anyone hasn't um added anything please feel free to give any last Reflections or questions arising so first as many people as would like can reflect that way um and then after every hand has been addressed then uh Ander if you could just give us a kind of look ahead so Avil first then anyone else with just where does this discussion take us and what do you want to continue to to learn more about FL oh uh so two things first if I walk out of here without understanding what is England I won't be disappointed and uh I I don't actually expect it to happen in like 12 hours that seems like it's something but more uh seriously I was a career I can I can agree with that difficulty that there is Unity between like physical structure and communication protocols but and that communication Protocols are quantum reference frame kind of individuate physical States individuate meaningful measurement outcome that our array file as physical possibilities but the question is then where do communication protocols slash Quantum reference frame come from like what are the physical determinants that bring about those things and uh that seems like a question that could be addressed at the conceptual level with this course and I would like uh I I saw the live stream 17. so that did not help uh anyway I would like to uh to other more grip into that if we can just say to the following courses thank you anyone else especially if they haven't added anything yet want to give kind of a closing yeah there's a lot to learn and Chris brings up a lot in his lecture even though we've only had one it's one of the denser hours epistemic resourcing that I've seen and there's also all of these accessory and Cutting Edge works that that Andre's showing um I'm really looking forward to how we can continue to curate and improve the questions that we have so that the uncertainties in our generative model that we have how would this be useful what does this mean why is this variable this way what is the implication for that those are all the great questions and let's just get them inscribed on our classical documents so that we can have other Quantum cognitive agents reduce their uncertainty by looking at the responses so with that I wonder if you could just look ahead to lecture two and just like Point towards where we're gonna be going uh yes so so lecture two will I think from what I hear from Chris is gonna start to is uh you know basic pathological aspects of the content measurement so the the two sources to look at would be two papers mainly this one physical meaning of the holographic principle I this one's by far the most pedagogical one so if I had to if I had to suggest that folks look at one it would be this one it's very easy to read I think uh but maybe for a bit more detail there's the slightly older one and they talk at length about corn cocoa and the high grounds here free energy principle for genetic content systems and so I think that's what's going to come in these two lectures basically the main ideas Behind These two papers Anya that's all I have to say all right well thank you for all who joined this first discussion hope that it was fun for you and for the audience check out the course site for the readings to get ahead of the second lecture we'll all watch the second lecture and come back in about a month for the second discussion everyone's welcome to join and the website has the registration information and so now that we've seen a little bit of one way it can be we may do something similar or we may do something different for the second discussion so thanks again everybody see you later